

gorch session report

Hardware: Apple M5

Date: 2026-04-30

Round: PR #11..#33 — perf round + plan 0001 Phase 1 primitives

1. Wall-clock improvements (before / after)

Numbers below are end-to-end measurements on the same hardware. “Before” = pre-session main; “After” = current main.

Benchmark	Before (ms)	After (ms)	Speedup
Encode (GPT-2 small, seq=64) — single sequence	55.70	28.70	1.94×
EncodeBatch16 (GPT-2 small, 16 seqs of 64 tokens, batched)	750.00	274.00	2.74×
GPT-2 generation (72 tokens, uncached)	35.50	37.20	0.95×
GPT-2 generation (72 tokens, KV cache)	—	3.10	12.00×
GPU autograd train step at GPT-2 dims (64, 768, 768)	2.27	0.32	7.10×

Encode (GPT-2 small, seq=64) — single sequence: PRs #11/#12/#14/#15: streaming safetensors, vectorised GELU, Linear alloc fix, NoGrad gating + scratch pool

EncodeBatch16 (GPT-2 small, 16 seqs of 64 tokens, batched): PR #12: batched encode + GELU vectorisation. The 'before' here is 16 sequential Encode calls on pre-session main.

GPT-2 generation (72 tokens, uncached): PR #16 baseline. Uncached path largely unchanged.

GPT-2 generation (72 tokens, KV cache): PR #5 + #16: KV cache integration. 12× faster than the uncached path, validates ADR-011.

GPU autograd train step at GPT-2 dims (64, 768, 768): PR #19: matmul size-threshold. Pre-fix gpt.ToMetal() was a 4.6× regression vs CPU; now at parity.

2. Mythos block per-component timings

Block config: mythos_tiny: dim=128, q-heads=4, kv-heads=2, experts=2/4, expert-dim=256

Sequence length: 64 tokens

Iterations per measurement: 30 (3 warmup excluded)

Component	Mean time (ms)	% of full block
RMSNorm.Forward	0.017	3.6%
GQA.Forward (with RoPE)	0.197	41.5%
MoE.Forward (4 experts, top-2)	0.286	60.3%
Full block (Norm+GQA+Add+Norm+MoE+Add)	0.473	
GPT-2 small Forward (reference)	31.729	

The mythos_tiny block (8.8M params, dim=128) runs at ~0.5 ms per forward call on M5 — about 60× faster per call than GPT-2 small (124M params, dim=768) at the same sequence length, which is the expected ratio of the model sizes.

3. Phase 1 primitives shipped

Plan 0001 Phase 1 calls for thirteen primitives. Twelve shipped in this round (the thirteenth — N-D permute — also landed). PRs:

- RMSNorm (#23)
- SiLU + SwiGLU (#27)
- Permute (#28)
- Gather + TopK + Multinomial + RepeatInterleave (#29)
- RoPE — Llama and GPT-NeoX styles (#30)
- GQA — Grouped-Query Attention (#31)
- MoE — Sparse Mixture of Experts FFN (#32)
- MLA — Multi-head Latent Attention (#33)
- AdamW + ClipGradNorm (#24)

Supporting changes

- Tensor.Detach() + NoGrad godoc warnings (#25)
- doc/plans/ lifted to main + 0004/0005 added (#22, #26)
- MoE/GQA/MLA composition smoke tests

4. Honest accounting — what's still missing

GPU autograd for non-matmul ops (LayerNorm/Softmax/GELU backward) — still CPU-only. Documented as ADR-009 deferred work; Plan 0004 has the implementation strategy. Closing this is the biggest remaining win for transformer training on Metal at shapes > 1G FMAs.

MoE expert weight training — MoE.Forward's manual scatter breaks the autograd chain on the FFN side. Router and per-expert Linear.Forward calls do produce autograd, but the weighted-sum scatter back into the output drops it. Needs a ScatterAdd primitive; filed as follow-up.

Softmax inside attention — softmaxInPlace operates on raw float32 and breaks autograd. Affects MultiHeadAttention's batched path, GQA, and MLA. Fixable by routing through public g.Softmax but ties into the FlashAttention-2 kernel work (plan 0004) where the whole attention is rewritten anyway.

bf16/fp16 dtype support — plan 0002. Largest single deferred perf win available (~2× memory + ~2× compute on Apple Silicon). Multi-week scope.

FlashAttention-2 fused Metal kernel — plan 0004. Eliminates the (seq, seq) intermediate that's 9 GB at GPT-2-12-layer × 4096 seq. 2-3 weeks of focused shader work.

MHA batched-path scale bug — divides by headDim instead of sqrt(headDim). Discovered while building GQA. Filed for a separate small PR; the per-head loop path is already correct.

5. Test summary

Unit tests: all packages green on the merged main (`go test ./...`).

End-to-end smoke tests (gated behind `-tags e2e`):

- **TestMythosBlockSmokeForward** — RMSNorm + GQA + RoPE + MoE composed in a transformer block, forward produces finite output that differs from input.
- **TestMythosBlockBackwardCompletes** — backward doesn't panic; documents the residual-only autograd state.
- **TestMythosBlockTrainStepConverges** — 30 AdamW steps drive MSE loss from ~6.5 to ~2.8 on a memorisation target.
- **TestFinetuneShortCorpusConverges** — real GPT-2 small fine-tune on a short sentence drops teacher-forced loss 5 orders of magnitude in 60 steps.

Numerical-vs-analytical autograd checks on every new differentiable op: RMSNorm, SiLU, SwiGLU (both inputs), Permute, Gather, RepeatInterleave, RoPE (Llama and GPT-NeoX styles), BatchedMatMul (dA and dB), BatchedMatMulTransB (dA and dB), Reshape.